

Diego Bautista Avilés

PhD Candidate in Physics

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Research Interests

Non-Hermitian Quantum Mechanics · Topological Phases of Matter · Floquet Systems · Chiral Phonons

Education

- Aug 2024–present **Doctor of Philosophy in Physics**, *Universidad de Chile*, Santiago, Chile
PhD Candidate (thesis proposal defended and approved). Supervisor: Prof. Luis E. F. Foà Torres. Doctoral Fellow, ANID. *Thesis: Topological Phases Induced by Exceptional Steering Protocols*. I engineer non-Hermitian topological phases in Floquet lattice models through protocols that alternate Hermitian and non-Hermitian dynamics (analogous to successive quantum measurements), targeting photonic-network emulators as experimental platforms.
- 2019–2023 **Bachelor of Science in Physics**, *Universidad Nacional Autónoma de México (UNAM)*, Mexico City, Mexico
GPA 9.95/10, ranked 1st out of 422 students. **Gabino Barreda Medal (UNAM)**: highest GPA of the Physics graduating class.
- 2016–2019 **High School (International Baccalaureate, graduated with honours)**, *Tecnológico de Monterrey (ITESM)*, Mexico

Research & Teaching Experience

- Aug 2024–present **Teaching Assistant**, *Universidad de Chile, Faculty of Physical and Mathematical Sciences*, Santiago, Chile
Courses: *Quantum Mechanics I* (Aug–Dec 2024; Aug–Dec 2025; Mar 2026–present), *Classical Mechanics* (Mar–May 2025), *Quantum Mechanics II* (Aug–Dec 2025; Mar 2026–present).
- Jun–Jul 2025 **Simons Visitor**, *International Centre for Theoretical Physics (ICTP)*, Trieste, Italy
I worked with Prof. Luis E. F. Foà Torres (Universidad de Chile).
- Jan–Feb 2025 **Research Visit**, *Karlsruhe Institute of Technology (KIT)*, Karlsruhe, Germany
I worked in the group of Prof. Alexander Mirlin.
- Jun–Jul 2024 **Simons Visitor**, *International Centre for Theoretical Physics (ICTP)*, Trieste, Italy
I worked with Prof. Luis E. F. Foà Torres (Universidad de Chile).
- Aug–Dec 2023 **Teaching Assistant, Atomic Physics & Condensed Matter**, *UNAM, Faculty of Sciences*, Mexico City, Mexico
- Jun–Jul 2023 **Summer Research Internship**, *University of Vienna*, Vienna, Austria
I was awarded a research contract as part of the Vienna Doctoral School of Physics program. Working under the supervision of Prof. Dr. Toma Susi (Physics of Nanostructured Materials Group), I extended the open-source abTEM STEM-simulation library by integrating DFT-derived (GPAW) inter-ionic electrostatic potentials to account for bonding effects in ionic materials neglected by the existing implementation, and designed systematic benchmarks evaluating their impact on simulated images.
- Mar 2021–May 2024 **Research Fellow**, *UNAM, Faculty of Chemistry*, Mexico City, Mexico
I worked under the supervision of Prof. Dr. José Eduardo Barrios Vargas, in collaboration with Prof. Luis E. F. Foà Torres (Universidad de Chile). I developed numerical codes to model phonon chirality in carbon nanotubes and helical structures, implementing pseudoangular momentum and Hausdorff chirality measures. I identified C₃-symmetric structures as optimal platforms for chiral phonons and ruled out carbon nanotubes as candidates, presenting results at multiple national and international conferences. *Dec 2023–May 2024: additionally affiliated with the CONAHCYT Frontier Research Project.*

Preprints

- [1] L. E. F. Foà Torres, G. Pappas, V. Achilleos, and **D. Bautista**, “Precision-Induced Irreversibility in non-Hermitian systems,” *arXiv preprint (submitted)*, 2025. arXiv: 2603.22284.
- [2] G. Pappas, **D. Bautista**, L. E. F. Foà Torres, and V. Achilleos, “Universal critical timescales in slow non-Hermitian dynamics,” *arXiv preprint (submitted)*, 2025. arXiv: 2604.01918.

Skills

Languages Spanish (Native), English (Advanced, TOEFL certified)
Programming Python, Mathematica, L^AT_EX, Java

Schools & Conferences

- Jul 2026 **XXVIII Giambiagi Winter School “New Directions in Condensed Matter”**, Buenos Aires, Argentina, Confirmed attendance
- Dec 2025 **X Reunión de Sólidos**, Bariloche, Argentina, Poster
Non-Hermitian Anomalous Floquet-Anderson Insulator.
- Nov 2025 **QUOST IX**, Valparaíso, Chile, Poster
Non-Hermitian Anomalous Floquet-Anderson Insulator.
- Sep–Oct 2025 **XXIII B-RMS Meeting**, Salvador, Brazil, Poster
Exploring Chiral Phonons in Simple Lattice Models.
- May 2025 **Joint ICTP-Lincei Conference on Quantum Physics: from Foundations to Emerging Technologies**, ICTP, Trieste, Italy, Attendee
- Nov 2023 **Gathering on Transport at Nanoscale**, Cuernavaca, Mexico, Oral presentation
Phonon Pseudo Angular Momentum in Graphene Structures.
- Oct 2023 **Congreso Nacional de Física**, Morelia, Mexico, Poster
Phonon Pseudo Angular Momentum in Graphene Structures.
- Mar 2023 **Conference and Advanced School on Low-Dimensional Quantum Systems (ICTP)**, Santiago, Chile, Attendee
- Nov 2022 **Reunión Mexicana de Físico-Química Teórica**, Cuernavaca, Mexico, Poster
Directional Control of Electric Current in a Ring-Shaped Device Partially Covered with a Graphene Bilayer.

Awards & Honours

- 2023 **Gabino Barreda Medal, UNAM**
Highest GPA of the Physics graduating class.
- 2016–2019 **High Academic Achievement Scholarship, ITESM**
- 2018 **Silver Medal, Mathematics Olympiad, Mexico City**

References

Prof. Luis E. F. Foà Torres Professor Department of Physics, Universidad de Chile luis.foatorres@uchile.cl	Prof. José Eduardo Barrios Associate Professor Dept. of Physical and Theoretical Chemistry, UNAM j.e.barrios@gmail.com
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